

Mathematics

Stage 9

Paper 1

2023

Cambridge Lower Secondary Progression Test

Name

Class

Date

1 hour

Additional materials: Geometrical instruments
Tracing paper (optional)

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.
- You are **not** allowed to use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 Tick (✓) to show which of these expressions is the number 87 000 000 written in standard form.

870×10^5 ☐

87×10^6 ☐

8.7×10^7 ☐

0.87×10^8 ☐

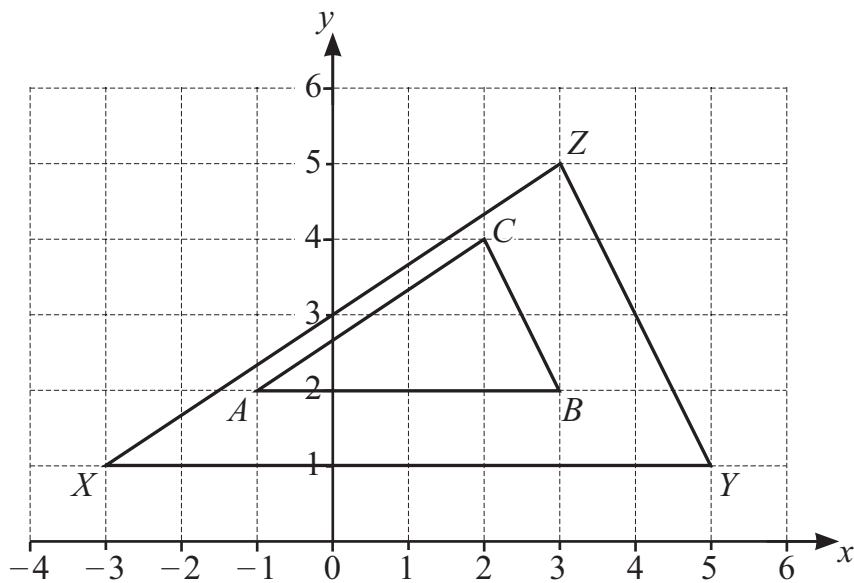
[1]

- 2 Complete the table of values for $y = x^2 + 4$

x	-2	-1	0	1	2
y		5	4		8

[1]

- 3 Here are two triangles, ABC and XYZ .



Describe fully the enlargement that maps triangle ABC onto triangle XYZ .

.....
 [2]

- 4 Here are two inequality symbols.



Write the correct inequality symbol in each box to complete these statements.

$$12.62 \times 0.91 \quad \boxed{\phantom{<}} \quad 12.62$$

$$12.62 \div 0.91 \quad \boxed{\phantom{<}} \quad 12.62$$

$$0.91 \div 12.62 \quad \boxed{\phantom{<}} \quad 12.62$$

[1]

- 5 In a game a player either wins, draws or loses.
 Gabriella plays the game.
 The probability that Gabriella wins is 0.55 and the probability that she draws is 0.15

Find the probability that Gabriella draws or loses her next game.

..... [2]

6 Here is an algebraic fraction.

$$\frac{12x + 4}{4}$$

Tick (✓) to show which expression is equivalent to this fraction.

$12x$

☐

$3x$

☐

$12x + 1$

☐

$3x + 1$

☐

$3x + 4$

☐

[1]

7 Here are the first four terms in a sequence.

$1, \quad 8, \quad 27, \quad 64$

(a) Find the next term in this sequence.

..... [1]

(b) Find the n th term of this sequence.

..... [1]

8 Given that $8.95 \times 5.62 = 50.299$ complete these calculations.

$$895 \times 0.562 = \dots\dots\dots$$

$$502.99 \div 56.2 = \dots\dots\dots$$

[2]

9 (a) Find the **exterior** angle of a regular pentagon.

$\dots\dots\dots^\circ$ [1]

(b) Find the **interior** angle of a regular pentagon.

$\dots\dots\dots^\circ$ [1]

(c) Tick (✓) to show which of these regular polygons will tessellate and which will not tessellate.

One has been done for you.

Regular polygons	Will tessellate	Will not tessellate
Equilateral triangles		
Squares	✓	
Regular pentagons		
Regular hexagons		

[1]

- 10 Eleven students each take a physics test and a chemistry test.

Mike starts to draw this back-to-back stem-and-leaf diagram showing the marks from the chemistry test.

Physics test					Chemistry test			
				4	9			
				5	2	8		
				6	1	1	7	8
				7	0	6	9	
				8	2			

Key: 2 | 4 | 9 represents 42 marks in physics and 49 marks in chemistry.

Here are the marks from the physics test.

64	73	60	55	46	42
69	88	55	71	57	

- (a) Complete the back-to-back stem-and-leaf diagram.

[2]

- (b) Complete the table.

	Physics	Chemistry
Mean	61.8	65.7
Median	60	
Mode	55	61
Range	46	33

[1]

(c) Tick (✓) to show which subject has marks that have a greater spread.

Physics

☐

Chemistry

☐

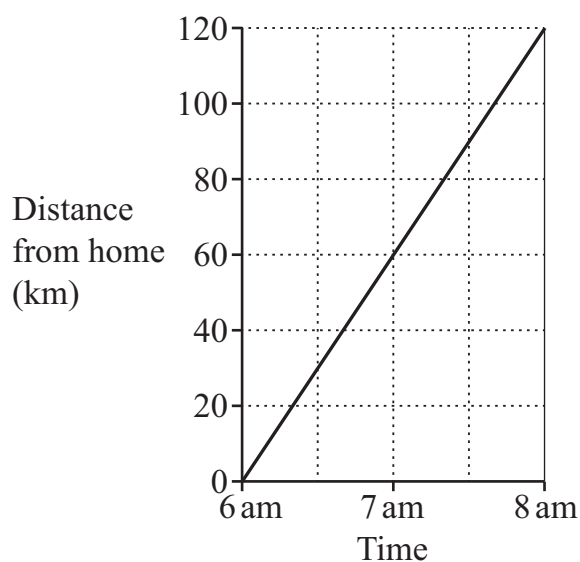
Explain how you know using appropriate values from the table in part (b).

.....

.....

[1]

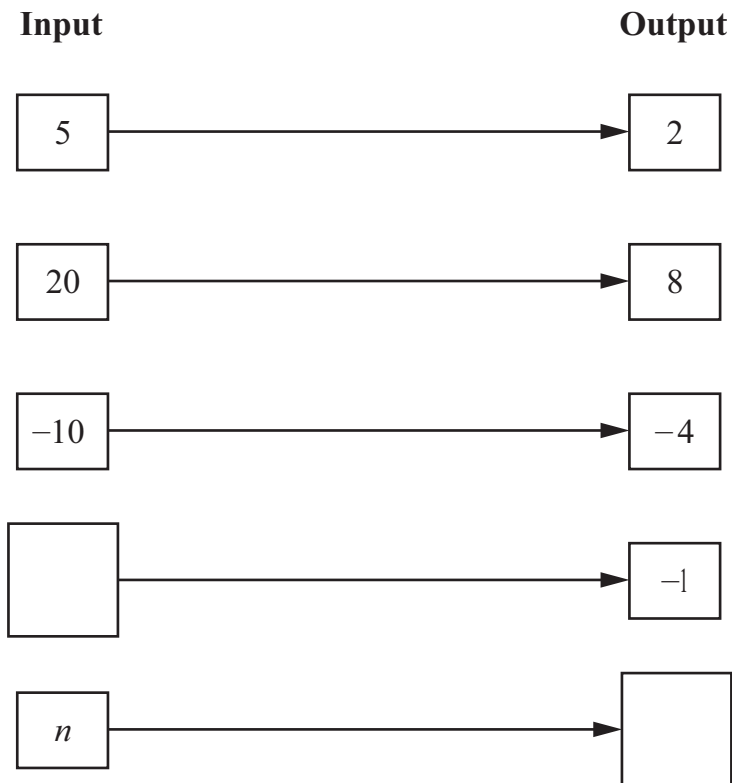
11 Here is the distance–time graph for a journey.



Calculate the speed for this journey.
Give your answer in km/h.

..... km/h [1]

12 Here are some inputs and outputs of the same function machine.



Complete the missing input and the missing output.

[2]

13 A speed camera records the speeds of 50 cars in one hour.

Speed, s (km/h)	Number of cars
$20 \leq s < 30$	9
$30 \leq s < 40$	17
$40 \leq s < 50$	18
$50 \leq s < 60$	4
$60 \leq s < 70$	2

(a) Write down the class interval that contains the median speed.

..... $\leq s <$ [1]

(b) Draw a ring around the value that **could** be the range of the 50 speeds.

10 16 18 48 70

[1]

- 14 When $n = -1$ **three** of these expressions have the same value.

$$2 - n^2$$

$$1 - 2n$$

$$n^2$$

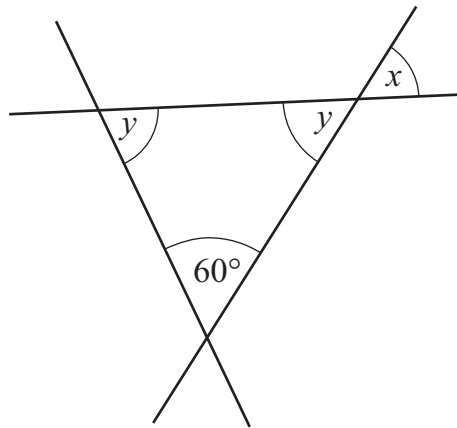
$$4n + 5$$

$$\frac{n}{2} + \frac{1}{2}$$

Draw a ring around **each** of the three expressions.

[2]

- 15 The diagram shows three straight lines crossing each other. Some of the angles are marked.



NOT TO
SCALE

Explain why angle x is 60° .
Give geometrical reasons in your answer.

.....

.....

..... [2]

16 The equation of a line is $5x + 2y = 6$

Find the gradient and the y -intercept of this line.

gradient =

y -intercept =

[3]

17 Work out.

$$1\frac{5}{6} + 2\frac{2}{3} \times \frac{3}{16}$$

Give your answer as a mixed number in its simplest form.

..... [4]

18 (a) Naomi thinks of a number n .

She changes her number using this rule

- multiply by 3
- then square
- then add 4

Tick (✓) to show the expression for her number after using this rule.

$$9n + 4 \quad \square \quad 3n^2 + 4 \quad \square \quad 9n^2 + 4 \quad \square \quad 3(n + 4)^2 \quad \square$$

[1]

(b) Mia thinks of a number m .

She changes her number using the rule $4(m - 5)^2$

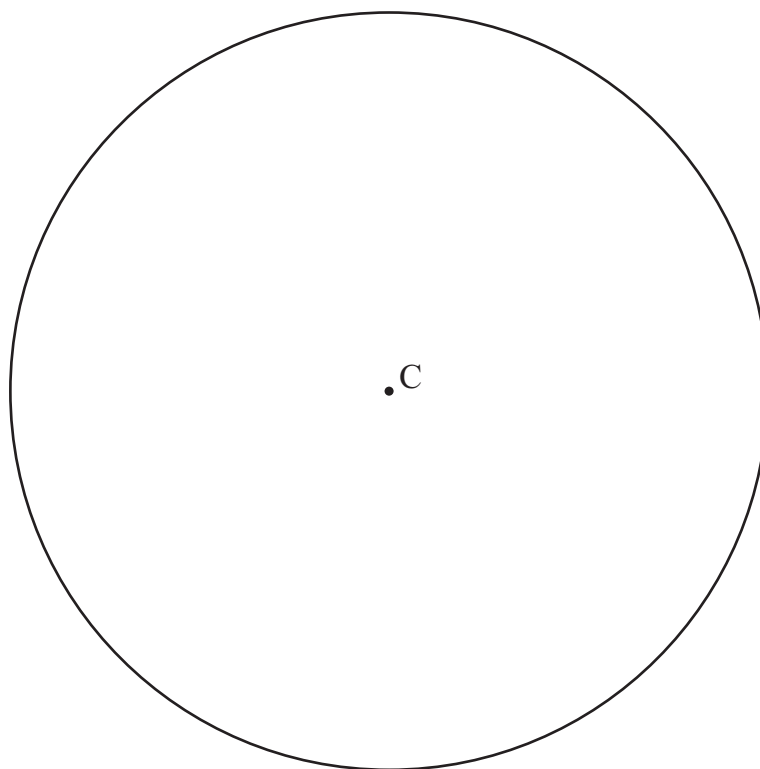
Complete the rule for Mia's number.

Her rule is

-
- then
- then

[1]

- 19 Here is a circle with centre C.



Use a straight edge and compasses only to construct an inscribed square.
Do **not** rub out your construction arcs and lines.

[2]

- 20 A straight line joins the points $A(2, 1)$ and $B(8, 10)$.
The point $C(6, y)$ lies on the line AB .

Find the y -coordinate of C .

$y =$ [1]

21 (a) In this question x is a **whole** number.

Find the **largest possible** value of x for which the inequality is correct.

$$\frac{27}{x} > \frac{x}{27}$$

$x =$ [1]

(b) In this question y is a **whole** number.

Find the **largest possible** value of y for which the inequality is correct.

$$\frac{2}{y} > \frac{y}{8}$$

$y =$ [1]

- 22 Yuri and Chen are playing a game.
They each have four cards numbered 1 to 4



They place their cards face down.
Each child turns over one of his cards at random.
The winner is the child who turns over the largest number.
If they turn over the same number, then neither of them wins.

Show that the probability that Yuri wins the game is $\frac{3}{8}$

[2]

23 Here are some equations involving indices.

$$(2^3)^2 \div 2^{-2} = 2^w$$

$$4^{-2} \times 4 \times 4^x = 4^{12}$$

$$6^y \times 6^y = 6^{-16}$$

Find the values of w , x and y .

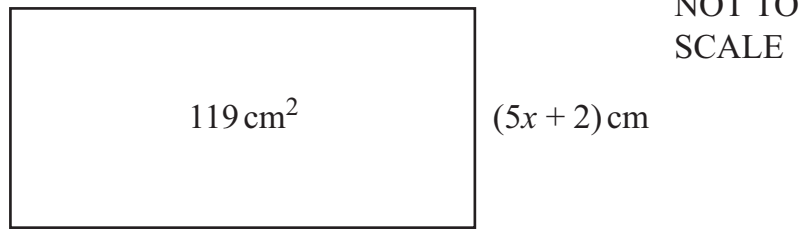
$w =$

$x =$

$y =$

[3]

24 Here is a rectangle.



The area of the rectangle is 119 cm^2 .

The length of the shorter side of the rectangle is $(5x + 2) \text{ cm}$.

- (a) Write down an expression in terms of x for the length of the longer side of the rectangle.

..... cm [1]

- (b) The longer side of the rectangle is 7 cm.

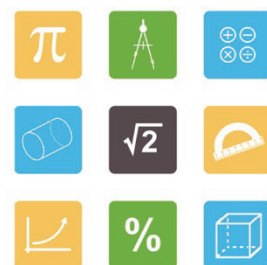
Find the value of x .

$x =$ [2]

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Mathematics

Stage 9

Paper 1

2023

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidance on marking**Difference in printing**

It is suggested that schools check their printed copies for differences in printing that may affect the answers to the questions, for example in measurement questions.

Brackets in mark scheme

When brackets appear in the mark scheme this indicates extra information that is not required but may be given.

For example:

Question	Answer	Mark	Part marks	Guidance
5	19.7 or 19.6(58...)	1		

This means that 19.6 is an acceptable truncated answer even though it is not the correct rounded answer.

The ... means you can ignore any numbers that follow this; you do not need to check them.

Accept

- any correct rounding of the numbers in the brackets, e.g. 19.66
- truncations beyond the brackets, e.g. 19.65

Do not accept

- 19.68 (since the numbers in brackets do not have to be present but if they are they should be correct).

These tables give general guidelines on marking learner responses that are not specifically mentioned in the mark scheme. Any guidance specifically given in the mark scheme supersedes this guidance.

Number and place value

The table shows various general rules in terms of acceptable decimal answers.

Accept
Accept omission of leading zero if answer is clearly shown, e.g. .675
Accept trailing zeros, unless the question has asked for a specific number of decimal places or significant figures, e.g. 0.7000
Accept a comma as a decimal point if that is the convention that you have taught the learners, e.g. 0,638

Units

For questions involving quantities, e.g. length, mass, money, duration or time, correct units must be given in the answer. Units are provided on the answer line unless finding the units is part of what is being assessed.

The table shows acceptable and unacceptable versions of the answer 1.85 m.

	Accept	Do not accept
If the unit is given on the answer line, e.g. m	Correct conversions, provided the unit is stated unambiguously, e.g.185 cm..... m (this is unambiguous since the unit cm comes straight after the answer, voiding the m which is now not next to the answer)	185..... m1850..... m etc.
If the question states the unit that the answer should be given in, e.g. 'Give your answer in metres'	1.85 1 m 85 cm	185; 1850 Any conversions to other units, e.g. 185 cm

Money

In addition to the rules for units, the table below gives guidance for answers involving money. The table shows acceptable and unacceptable versions of the answer \$0.30

	Accept	Do not accept
If the amount is in dollars and cents, the answer should be given to two decimal places	\$0.30 For an integer number of dollars it is acceptable not to give any decimal places, e.g. \$9 or \$9.00	\$0.3 \$09 or \$09.00
If units are not given on the answer line	Any unambiguous indication of the correct amount, e.g. 30 cents; 30 c \$0.30; \$0-30; \$0=30; \$00:30	30 or 0.30 without a unit \$30; 0.30 cents Ambiguous answers, e.g. \$30 cents; \$0.30 c; \$0.30 cents (as you do not know which unit applies because there are units either side of the number)
If \$ is shown on the answer line	All unambiguous indications, e.g. \$.....0.30.....; \$.....0-30.....; \$.....0=30.....; \$.....00:30.....	\$.....30..... Ambiguous answers, e.g. \$.....30 cents.....; \$.....0.30 cents..... unless units on the answer line have been deleted, e.g. \$.....30 cents.....
If cents is shown on the answer line	30.....cents	0.30.....cents Ambiguous answers, e.g.\$30cents;\$0.30cents unless units on the answer line have been deleted, e.g.\$0.30.....cents

Duration

In addition to the rules for units, the table below gives guidance for answers involving time durations. The table shows acceptable and unacceptable versions of the answer 2 hours and 30 minutes.

Accept	Do not accept
Any unambiguous indication using any reasonable abbreviations of hours (h, hr, hrs), minutes (m, min, mins) and seconds (s, sec, secs), e.g. 2 hours 30 minutes; 2 h 30 m; 02 h 30 m	Incorrect or ambiguous formats, e.g. 2.30; 2.3; 2.30 hours; 2.30 min; 2 h 3; 2.3 h (this is because this indicates 0.3 of an hour (i.e. 18 minutes) rather than 30 minutes)
Any correct conversion with appropriate units, e.g. 2.5 hours; 150 mins unless the question specifically asks for time given in hours and minutes	02:30 (as this is a 24-hour clock time, not a time interval) 2.5; 150

Time

The table below gives guidance for answers involving time.

The table shows acceptable and unacceptable versions of the answer 07:30

	Accept	Do not accept
If the answer is required in 24-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 07:30 with any separator in place of the colon, e.g. 07 30; 07,30; 07-30; 0730	7:30 7:30 am 7 h 30 m 7:3 730 7.30 pm 073 07.3
If the answer is required in 12-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 7:30 am with any separator in place of the colon, e.g. 7 30 am; 7.30 am; 7-30 am 7.30 in the morning Half past seven (o'clock) in the morning Accept am or a.m.	Absence of am or pm 1930 am 7 h 30 m 7:3 730 7.30 pm

Algebra

The table shows acceptable and unacceptable versions of the answer $3x - 2$

Accept	Do not accept
$x^3 - 2$; $3 \times x - 2$	$3x + -2$ if it is supposed to be in simplest form
Case change in letters	
Changes in letters as long as there is no ambiguity	

Accept extra brackets when factorising, e.g. $5(x + (3 + y))$

Teachers must mark the final answer given. If a correct answer is seen in working but final answer is given incorrectly then the final answer must be marked. If no answer is given on the answer line then the final line of the working can be taken to be the final answer.

Inequalities

The table shows acceptable and unacceptable versions of various answers.

For the following	Accept	Do not accept
For $6 \leq x < 8$	$[6, 8)$	$< x <$
For $x \leq -2$	$(-\infty, -2]$	$x < -2$
For $x > 3$	$(3, \infty)$ $3 < x$	Just '3' written on the answer line, even if $x > 3$ appears in the working

Plotting points

The table shows acceptable and unacceptable ways to plot points.

Accept	Do not accept
Crosses or dots plotted within $\pm \frac{1}{2}$ square of the correct answer The graph line passing through a point implies the point even though there is no cross	A horizontal line and vertical line from the axes meeting at the required point

Question	Answer						Marks	Part Marks	Guidance
1	8.7×10^7						1		Accept any clear indication.
2	x	(−2)	(−1)	(0)	(1)	(2)	1		Both answers correct for the mark.
	y	8	(5)	(4)	5	(8)			
3	Scale factor of 2 and Centre (1, 3)						2	Award 1 mark for Scale factor of 2 or Centre (1, 3)	
4	< > <						1		All three answers correct for the mark.
5	0.45						2	Award 1 mark for 1 − 0.55 or (1 − 0.15 − 0.55) + 0.15 or P(loses) = 0.3	
6	$3x + 1$						1		Accept any clear indication.

Question	Answer	Marks	Part Marks	Guidance															
7(a)	125	1																	
7(b)	n^3	1		Accept equivalents, e.g. $n \times n \times n$															
8	502.99 and 8.95	2	Award 1 mark for 502.99 or 8.95																
9(a)	72(°)	1																	
9(b)	108(°)	1		Accept follow through from 180 – <i>their (a)</i> .															
9(c)	<table><tr><th>Regular polygons</th><th>Will tessellate</th><th>Will not tessellate</th></tr><tr><td>Equilateral triangles</td><td>✓</td><td></td></tr><tr><td>Squares</td><td>(✓)</td><td></td></tr><tr><td>Regular pentagons</td><td></td><td>✓</td></tr><tr><td>Regular hexagons</td><td>✓</td><td></td></tr></table>	Regular polygons	Will tessellate	Will not tessellate	Equilateral triangles	✓		Squares	(✓)		Regular pentagons		✓	Regular hexagons	✓		1		All three answers correct for the mark. Accept any clear indication.
Regular polygons	Will tessellate	Will not tessellate																	
Equilateral triangles	✓																		
Squares	(✓)																		
Regular pentagons		✓																	
Regular hexagons	✓																		

Question	Answer	Marks	Part Marks	Guidance																									
10(a)	Physics test <table border="1"> <tr> <td></td><td></td><td>6</td><td>2</td><td>(4)</td></tr> <tr> <td></td><td>7</td><td>5</td><td>5</td><td>(5)</td></tr> <tr> <td></td><td>9</td><td>4</td><td>0</td><td>(6)</td></tr> <tr> <td></td><td></td><td>3</td><td>1</td><td>(7)</td></tr> <tr> <td></td><td></td><td></td><td>8</td><td>(8)</td></tr> </table>			6	2	(4)		7	5	5	(5)		9	4	0	(6)			3	1	(7)				8	(8)	2	Award 1 mark for correct unordered leaves or for three correct rows.	Award 2 marks for fully correct drawing with ordered leaves.
		6	2	(4)																									
	7	5	5	(5)																									
	9	4	0	(6)																									
		3	1	(7)																									
			8	(8)																									
10(b)	67	1																											
10(c)	Physics ticked and A correct explanation, e.g. the range for physics is 46 which is bigger (than the range for chemistry which is 33).	1		Accept 46 is bigger than 33																									
11	60 (km/h)	1																											
12	$-\frac{5}{2}$ or -2.5 and $\frac{2n}{5}$ or $0.4n$	2	Award 1 mark for $-\frac{5}{2}$ or -2.5 or $\frac{2n}{5}$ or $0.4n$	Accept equivalents, e.g. $n \div 5 \times 2$ for $\frac{2n}{5}$																									

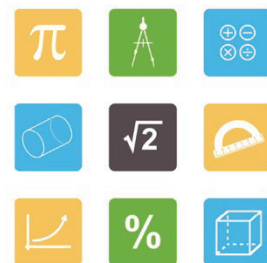
Question	Answer	Marks	Part Marks	Guidance
13(a)	$30 \leq s < 40$	1		Both values correct for the mark.
13(b)	48	1		Accept any clear indication.
14	$2 - n^2$ and n^2 and $4n + 5$	2	Award 1 mark for two correct answers and no more than one incorrect answer.	Accept any clear indication.
15	The angles in a triangle sum to $180(^{\circ})$ or the triangle must be equilateral. and Vertically opposite angles are equal.	2	Award 1 mark for The angles in a triangle sum to $180(^{\circ})$ or the triangle must be equilateral or Vertically opposite angles are equal.	Accept the triangle is an isosceles triangle and $y = 60$ Accept opposite angles are equal. An incorrect statement invalidates a correct statement.
16	(gradient =) $-\frac{5}{2}$ or equivalent and (y-intercept =) 3	3	Award 2 marks for one answer correct. or Award 1 mark for correct rearrangement of the line in the form $y = -\frac{5}{2}x + 3$ or for both correct if following through from <i>their</i> equation in the form $y = mx + c$	Accept in the form $y = 3 - \frac{5}{2}x$ for 1 mark.

Question	Answer	Marks	Part Marks	Guidance
17	$2\frac{1}{3}$	4	Award 3 marks for $\frac{7}{3}$ or equivalent. or Award 2 marks for $\frac{8}{3} \times \frac{3}{16} = \frac{1}{2}$ or equivalent. or Award 1 mark for $2\frac{2}{3} = \frac{8}{3}$ or $1\frac{5}{6} = \frac{11}{6}$ or $1\frac{5}{6} + \frac{k}{6}$	For 3 marks, accept equivalent answers, e.g. $1\frac{8}{6}$, $2\frac{2}{6}$ For 1 mark, accept any correct use of a common denominator with <i>their</i> $\frac{1}{2}$
18(a)	$9n^2 + 4$	1		Accept any clear indication.
18(b)	Subtract 5 (then) square (then) multiply by 4	1		Accept 'take 5' or '- 5' Accept 'times 4' or '× 4'
19	Correct square with diameter of the circle shown as a diagonal of the square and Arcs for the perpendicular bisector on the other diagonal.	2	Award 1 mark for Correct square with some/all construction arcs missing or Diameter and perpendicular diameter correct with construction arcs drawn.	
20	(y =) 7	1		

Question	Answer	Marks	Part Marks	Guidance																																	
21(a)	(x =) 26	1																																			
21(b)	(y =) 3	1																																			
22	Convincing work leading to $\frac{6}{16} = \frac{3}{8}$ e.g.16 outcomes and 6 ways in which Yuri wins indicated. <table><tr><td colspan="2"></td><td colspan="4">Yuri</td></tr><tr><td colspan="2"></td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td rowspan="4">Chen</td><td>1</td><td></td><td>*</td><td>*</td><td>*</td></tr><tr><td>2</td><td></td><td></td><td>*</td><td>*</td></tr><tr><td>3</td><td></td><td></td><td></td><td>*</td></tr><tr><td>4</td><td></td><td></td><td></td><td></td></tr></table> and $\frac{6}{16} = \frac{3}{8}$			Yuri						1	2	3	4	Chen	1		*	*	*	2			*	*	3				*	4					2	Award 1 mark for Listing the 6 ways that Yuri wins or A sample space diagram showing the 16 possible outcomes or Probability of any one outcome as $\frac{1}{16}$ or $\frac{1}{4} \times \frac{1}{4}$	Other examples of convincing work for 2 marks: The six ways that Yuri wins listed: 2, 1; 3, 1; 3, 2; 4, 1; 4, 2; 4, 3 and the probability of each as $\frac{1}{16}$ or $\frac{1}{4} \times \frac{1}{4}$ leading to $\frac{6}{16} = \frac{3}{8}$ or $\frac{1}{4} \times \frac{1}{4} + \frac{1}{4} \times \frac{2}{4} + \frac{1}{4} \times \frac{3}{4} = \frac{3}{8}$ or The four ways that neither of them wins considered with probability of each as $\frac{1}{16}$ or $\frac{1}{4} \times \frac{1}{4}$ leading to $\frac{1}{2} \left(1 - \frac{4}{16} \right) = \frac{3}{8}$
		Yuri																																			
		1	2	3	4																																
Chen	1		*	*	*																																
	2			*	*																																
	3				*																																
	4																																				
23	(w =) 8 (x =) 13 (y =) -8	3	Award 1 mark for each correct answer.																																		

Question	Answer	Marks	Part Marks	Guidance
24(a)	$\frac{119}{5x+2}$ (cm)	1		
24(b)	$(x =) 3$	2	Award 1 mark for $119 = 7(5x + 2)$ or better or for <i>their</i> expression from part (a) equated to 7 with one step towards solving for x .	Or better, e.g. $\frac{119}{7} = 5x + 2$ or $119 = 35x + 14$ or $105 = 35x$ etc.

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Mathematics

Stage 9

Paper 2

2023

Cambridge Lower Secondary Progression Test

Name

Class

Date

1 hour

Additional materials: Calculator
Geometrical instruments
Tracing paper (optional)

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 It takes 3 people 10 days to build a wall.

Find how many days it would take 6 people to build the same wall working at the same rate.

..... [1]

- 2 The term-to-term rule for a sequence is 'subtract k '.
The 1st term of the sequence is 40
The 4th term of the sequence is 35.5

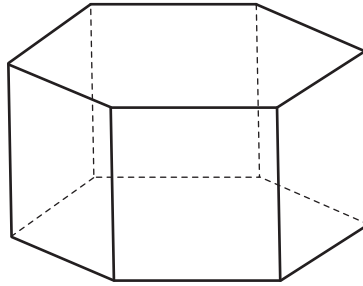
(a) Find the 2nd and 3rd terms of the sequence.

40, , , 35.5 [1]

(b) Find the first negative term in the sequence.

..... [1]

- 3 A prism has a cross-section that is a regular hexagon.



NOT TO
SCALE

- (a) Write down the number of planes of symmetry of this prism.

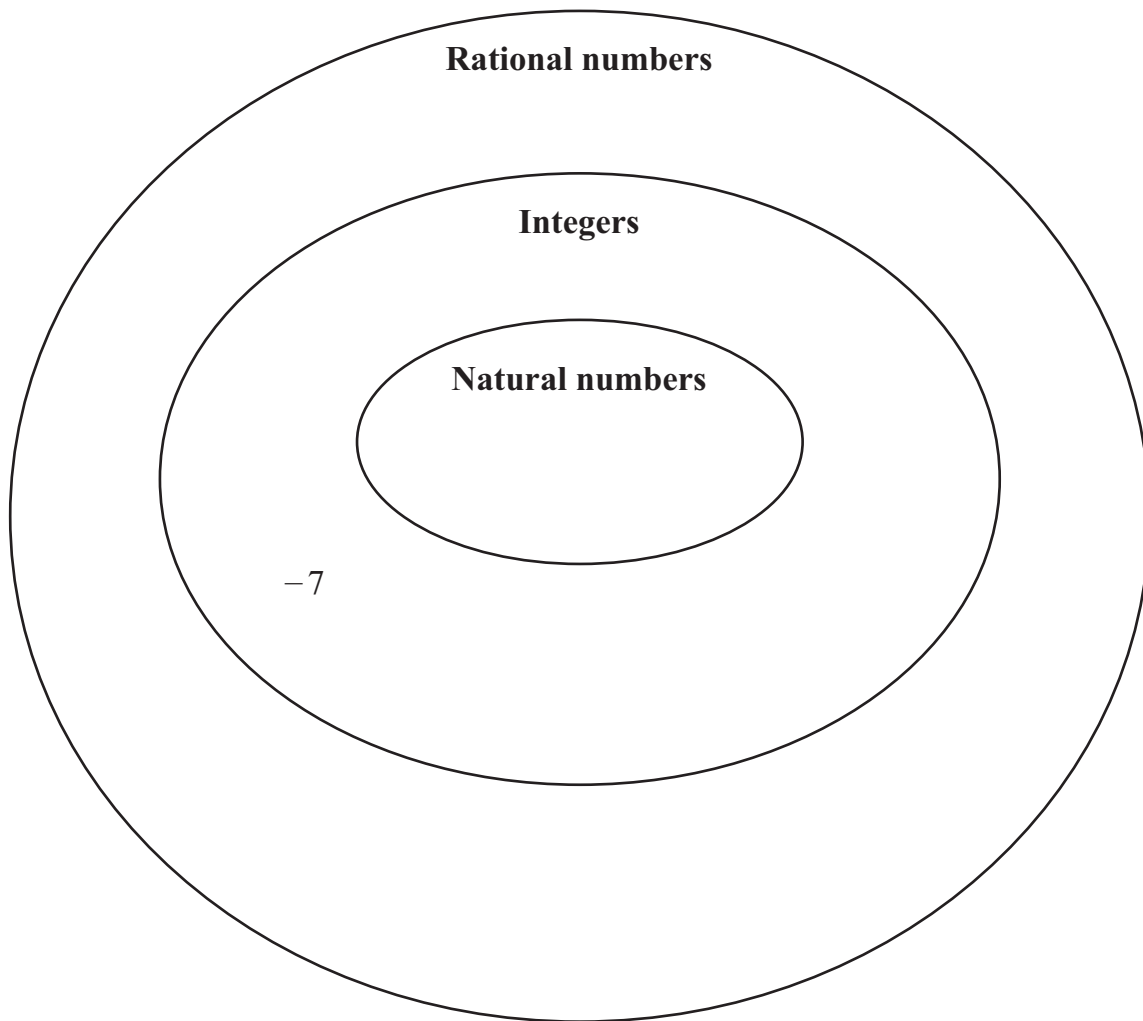
..... [1]

- (b) The area of the cross-section of the prism is 10.4 m^2 .
The volume of the prism is 8.84 m^3 .

Calculate the height of the prism.

..... m [1]

4 (a) Here is a Venn diagram.



Write each of these numbers in the correct part of the Venn diagram.
One has been done for you.

-7 12 $\sqrt{36}$ 4.7 $-\frac{55}{11}$ $-\frac{2}{3}$

[2]

(b) Here is a list of words.

rational irrational prime odd square

Complete the sentence using the correct word from the list.

$\sqrt{3}$ and $\sqrt{5}$ are examples of numbers.

[1]

5 Expand and simplify.

$$(x + 7)(x + 2)$$

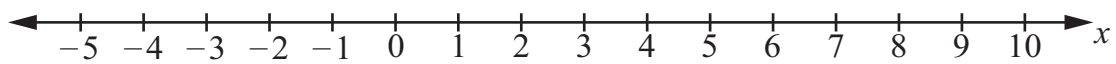
..... [2]

6 (a) Solve the inequality.

$$-2 \leq x - 3 < 5$$

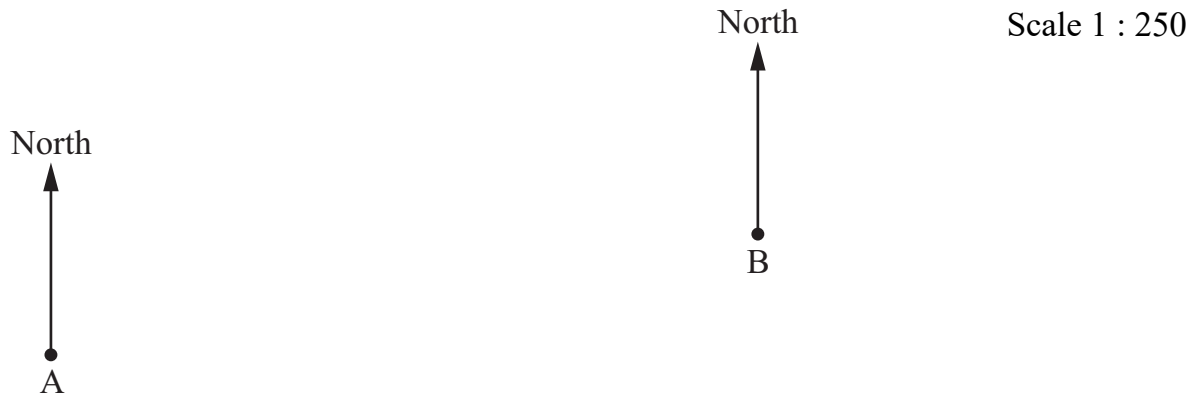
..... [1]

(b) Show your solution to part (a) on the number line.



[1]

- 7 The map shows the position of two trees, A and B.



Tree C is on a bearing of 110° from tree A.

Tree C is on a bearing of 205° from tree B.

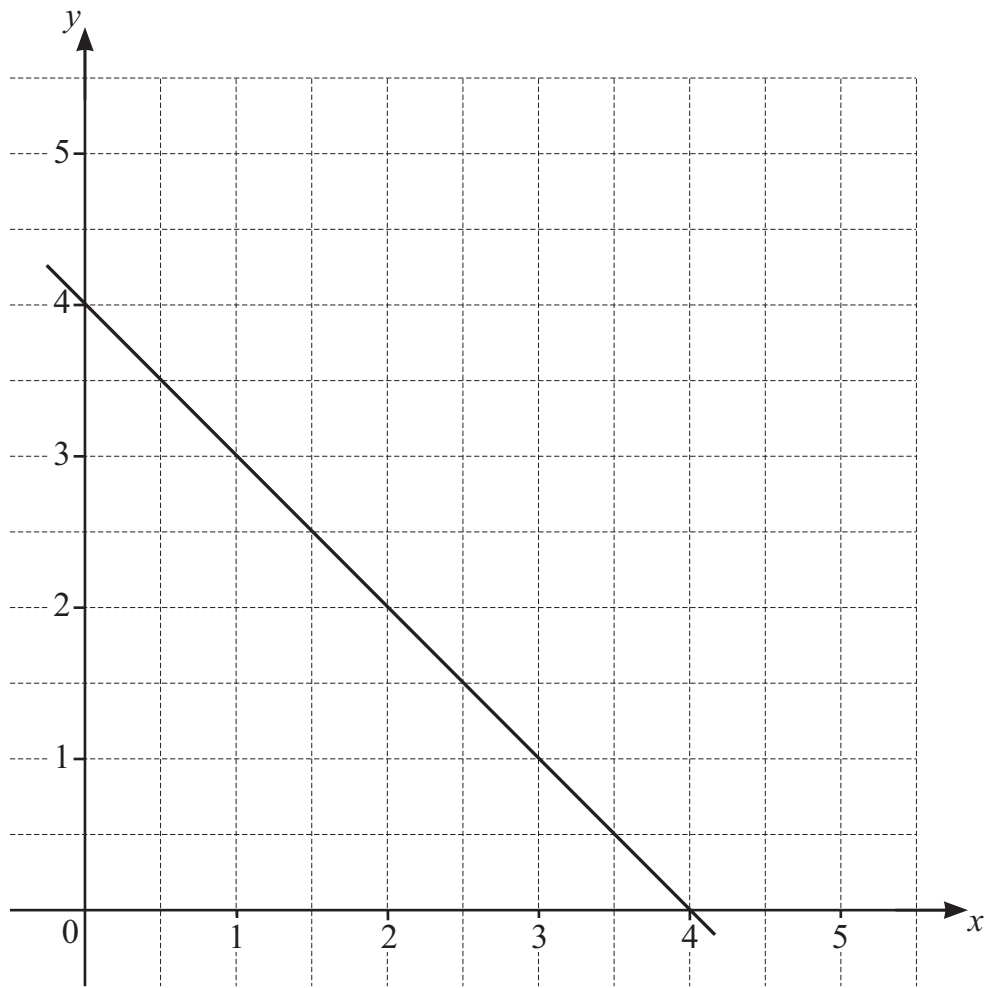
The scale of the map is 1 : 250

Find the actual distance, in metres, between tree B and tree C.

You should show the position of tree C on the map.

..... m [3]

- 8 The graph of $x + y = 4$ is drawn on the grid.



- (a) Draw the graph of $y - x = 1$ on the same grid.

[2]

- (b) Use the graph to write down the solution to the pair of simultaneous equations

$$\begin{aligned} x + y &= 4 \\ \text{and} \\ y - x &= 1 \end{aligned}$$

$x =$
 $y =$
 [1]

- 9 A teaspoon of sugar has a mass of 4.2 grams.

(a) Write down the mass of this sugar in milligrams.

..... mg [1]

(b) Mike says, 'There are 12 000 grains of sugar in one teaspoon.'

Use Mike's value to calculate the mass of one grain of sugar, **in grams**.
Give your answer in standard form.

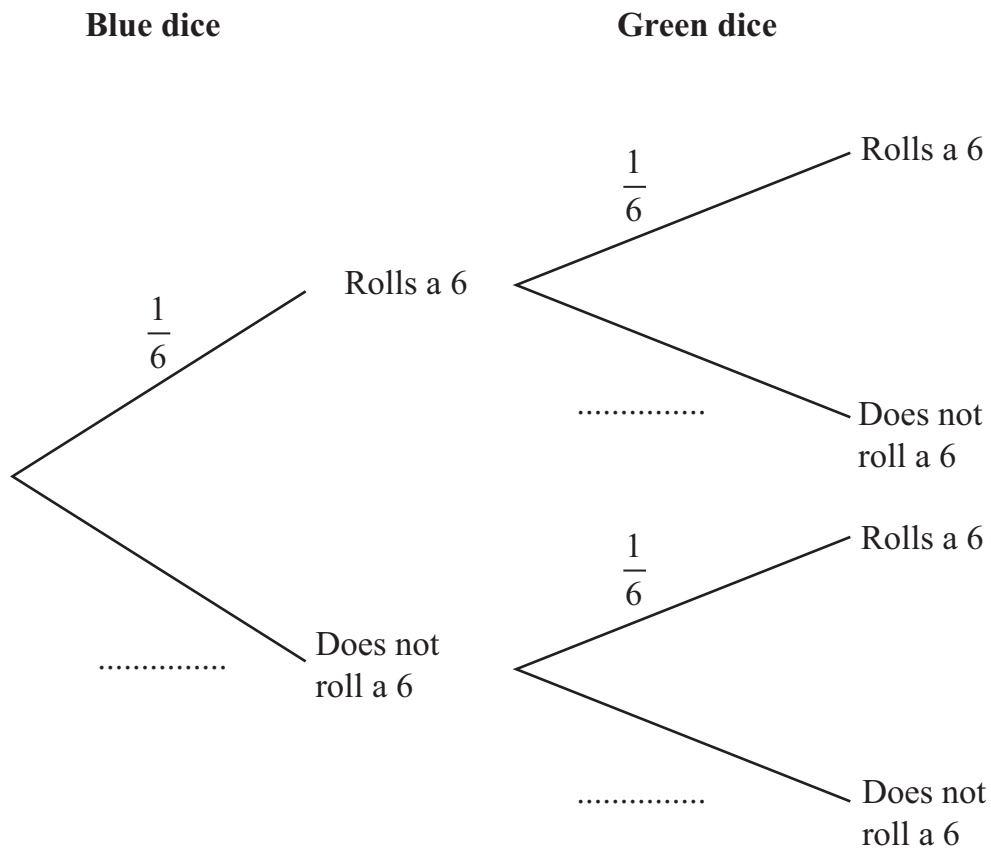
..... g [2]

- 10 Two friends share some money in the ratio 3 : 5
One friend gets \$1.50

Find the **two** possible amounts that the other friend gets.

\$
or \$
[2]

11 Safia rolls a fair blue dice and a fair green dice.



(a) Complete the tree diagram.

[1]

(b) Find the probability that she rolls exactly one 6

..... [2]

(c) Complete the sentence using a word or words from the list.

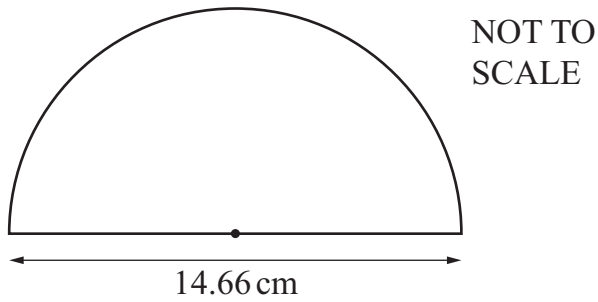
dependent independent mutually exclusive complementary

‘Safia rolls a 6 on the blue dice’ and ‘Safia does not roll a 6 on the green dice’ are

..... events.

[1]

12 Here is a semicircle.



Calculate the area of the semicircle.

Give your answer correct to **three** significant figures.

..... cm² [3]

13 An elk has a top speed of 72.4 kilometres per hour.
An ostrich has a top speed of 43.0 miles per hour.

Tick (✓) to show which of these animals has the fastest top speed.
You must show your working.

Elk ☐

Ostrich ☐

[2]

14 Samira wants to compare the heights of boys and girls when they are 14 years old.

- (a) She takes a random sample of five boys from the local running club and five girls from her class at school.

The children that she samples are all 14 years old.

Give two reasons why this may **not** be a good sample.

1

.....

2

.....

[2]

- (b) Samira measures the height, h cm, of each child.

Samira wants to record the heights using these class intervals.

$$h < 150$$

$$150 < h < 160$$

$$160 < h < 170$$

$$h > 170$$

Explain why she may **not** be able to record every child's height in these class intervals.

.....

..... [1]

- (c) Write down the correct class intervals that allow Samira to record every child's height. One has been done for you.

$$h < 150$$

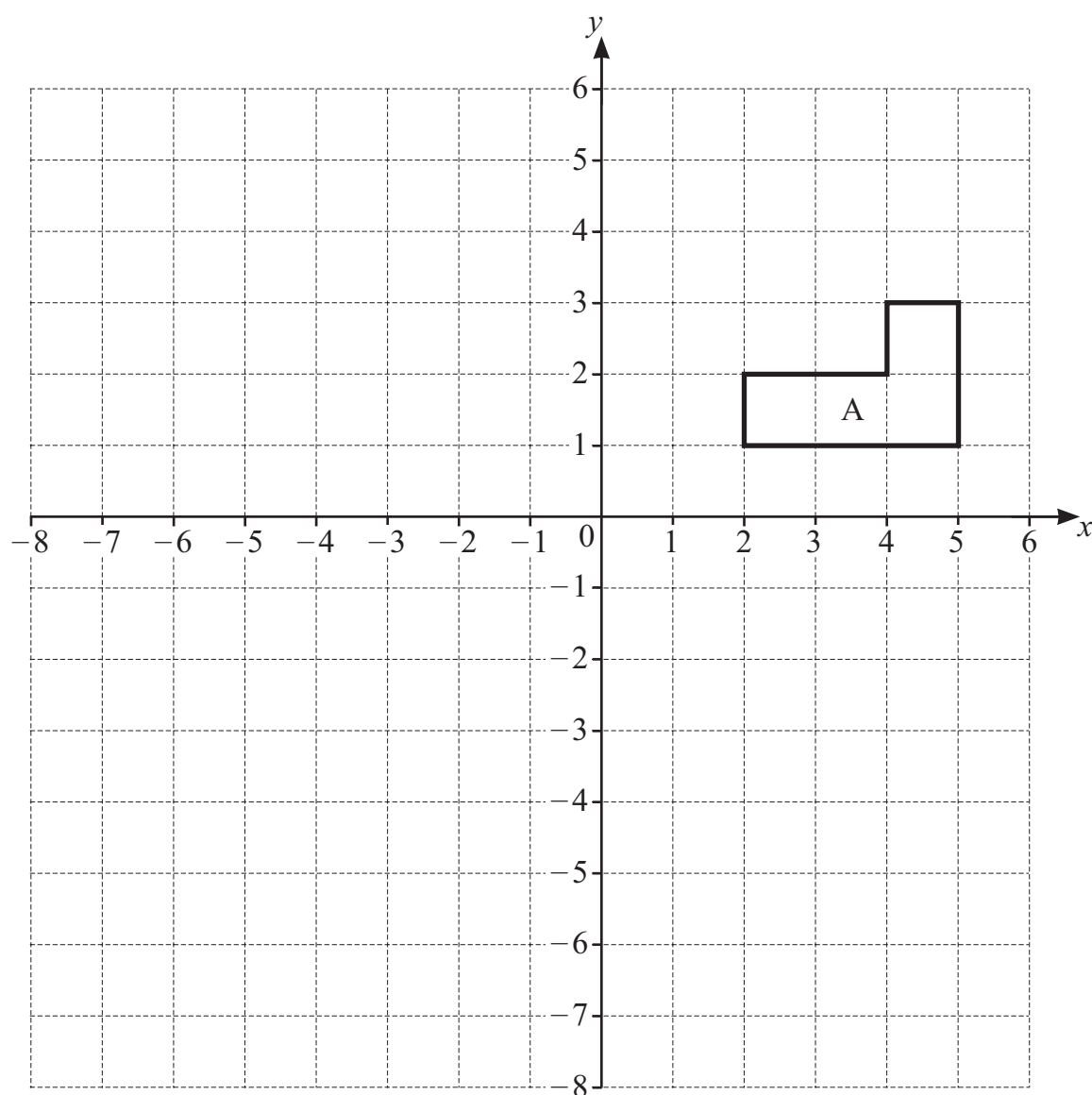
.....

.....

.....

[1]

15 Shape A is shown on a grid.



- (a) Reflect shape A in the line $x = 3$ and label the image B.
Then reflect shape B in the line $y = -2$ and label the image C.

[2]

- (b) Describe fully the **single** transformation that maps shape A onto shape C.

.....
..... [2]

16 A formula used in physics is $v^2 = u^2 + 2as$

(a) Find the **two** possible values of v when $u = 2.1$ $a = -9.8$ $s = -5.4$

$v =$

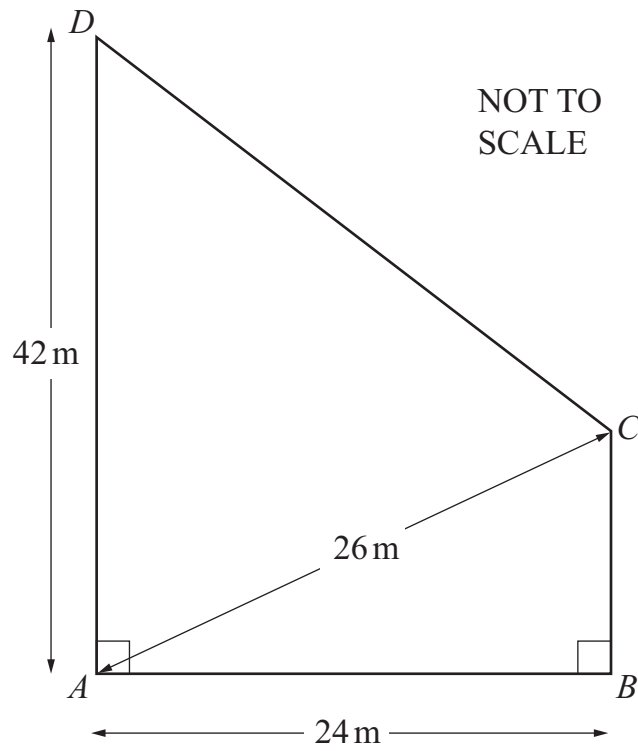
or $v =$

[2]

(b) Rearrange this formula to make a the subject.

$a =$ [2]

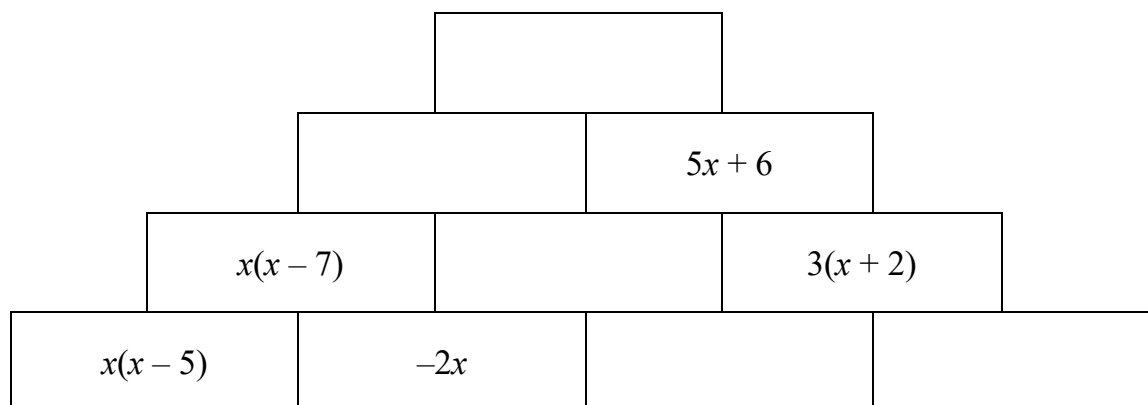
- 17 A field is in the shape of a trapezium $ABCD$.



Calculate the perimeter of the field.

..... m [3]

- 18** In this pyramid each expression is equal to the sum of the two expressions immediately below it.



Complete the pyramid, writing each expression in its simplest form.

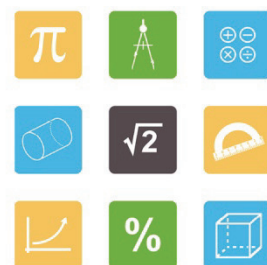
[3]

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Mathematics

Stage 9

Paper 2

2023

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidance on marking**Difference in printing**

It is suggested that schools check their printed copies for differences in printing that may affect the answers to the questions, for example in measurement questions.

Brackets in mark scheme

When brackets appear in the mark scheme this indicates extra information that is not required but may be given.

For example:

Question	Answer	Mark	Part marks	Guidance
5	19.7 or 19.6(58...)	1		

This means that 19.6 is an acceptable truncated answer even though it is not the correct rounded answer.

The ... means you can ignore any numbers that follow this; you do not need to check them.

Accept

- any correct rounding of the numbers in the brackets, e.g. 19.66
- truncations beyond the brackets, e.g. 19.65

Do not accept

- 19.68 (since the numbers in brackets do not have to be present but if they are they should be correct).

These tables give general guidelines on marking learner responses that are not specifically mentioned in the mark scheme. Any guidance specifically given in the mark scheme supersedes this guidance.

Number and place value

The table shows various general rules in terms of acceptable decimal answers.

Accept
Accept omission of leading zero if answer is clearly shown, e.g. .675
Accept trailing zeros, unless the question has asked for a specific number of decimal places or significant figures, e.g. 0.7000
Accept a comma as a decimal point if that is the convention that you have taught the learners, e.g. 0,638

Units

For questions involving quantities, e.g. length, mass, money, duration or time, correct units must be given in the answer. Units are provided on the answer line unless finding the units is part of what is being assessed.

The table shows acceptable and unacceptable versions of the answer 1.85 m.

	Accept	Do not accept
If the unit is given on the answer line, e.g. m	Correct conversions, provided the unit is stated unambiguously, e.g.185 cm..... m (this is unambiguous since the unit cm comes straight after the answer, voiding the m which is now not next to the answer)	185..... m1850..... m etc.
If the question states the unit that the answer should be given in, e.g. 'Give your answer in metres'	1.85 1 m 85 cm	185; 1850 Any conversions to other units, e.g. 185 cm

Money

In addition to the rules for units, the table below gives guidance for answers involving money. The table shows acceptable and unacceptable versions of the answer \$0.30

	Accept	Do not accept
If the amount is in dollars and cents, the answer should be given to two decimal places	\$0.30 For an integer number of dollars it is acceptable not to give any decimal places, e.g. \$9 or \$9.00	\$0.3 \$09 or \$09.00
If units are not given on the answer line	Any unambiguous indication of the correct amount, e.g. 30 cents; 30 c \$0.30; \$0-30; \$0=30; \$00:30	30 or 0.30 without a unit \$30; 0.30 cents Ambiguous answers, e.g. \$30 cents; \$0.30 c; \$0.30 cents (as you do not know which unit applies because there are units either side of the number)
If \$ is shown on the answer line	All unambiguous indications, e.g. \$.....0.30.....; \$.....0-30.....; \$.....0=30.....; \$.....00:30.....	\$.....30..... Ambiguous answers, e.g. \$.....30 cents.....; \$.....0.30 cents..... unless units on the answer line have been deleted, e.g. \$.....30 cents.....
If cents is shown on the answer line	30.....cents	0.30.....cents Ambiguous answers, e.g.\$30cents;\$0.30cents unless units on the answer line have been deleted, e.g.\$0.30.....cents

Duration

In addition to the rules for units, the table below gives guidance for answers involving time durations. The table shows acceptable and unacceptable versions of the answer 2 hours and 30 minutes.

Accept	Do not accept
Any unambiguous indication using any reasonable abbreviations of hours (h, hr, hrs), minutes (m, min, mins) and seconds (s, sec, secs), e.g. 2 hours 30 minutes; 2 h 30 m; 02 h 30 m	Incorrect or ambiguous formats, e.g. 2.30; 2.3; 2.30 hours; 2.30 min; 2 h 3; 2.3 h (this is because this indicates 0.3 of an hour (i.e. 18 minutes) rather than 30 minutes)
Any correct conversion with appropriate units, e.g. 2.5 hours; 150 mins unless the question specifically asks for time given in hours and minutes	02:30 (as this is a 24-hour clock time, not a time interval) 2.5; 150

Time

The table below gives guidance for answers involving time.

The table shows acceptable and unacceptable versions of the answer 07:30

	Accept	Do not accept
If the answer is required in 24-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 07:30 with any separator in place of the colon, e.g. 07 30; 07,30; 07-30; 0730	7:30 7:30 am 7 h 30 m 7:3 730 7.30 pm 073 07.3
If the answer is required in 12-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 7:30 am with any separator in place of the colon, e.g. 7 30 am; 7.30 am; 7-30 am 7.30 in the morning Half past seven (o'clock) in the morning Accept am or a.m.	Absence of am or pm 1930 am 7 h 30 m 7:3 730 7.30 pm

Algebra

The table shows acceptable and unacceptable versions of the answer $3x - 2$

Accept	Do not accept
$x^3 - 2$; $3 \times x - 2$	$3x + -2$ if it is supposed to be in simplest form
Case change in letters	
Changes in letters as long as there is no ambiguity	

Accept extra brackets when factorising, e.g. $5(x + (3 + y))$

Teachers must mark the final answer given. If a correct answer is seen in working but final answer is given incorrectly then the final answer must be marked. If no answer is given on the answer line then the final line of the working can be taken to be the final answer.

Inequalities

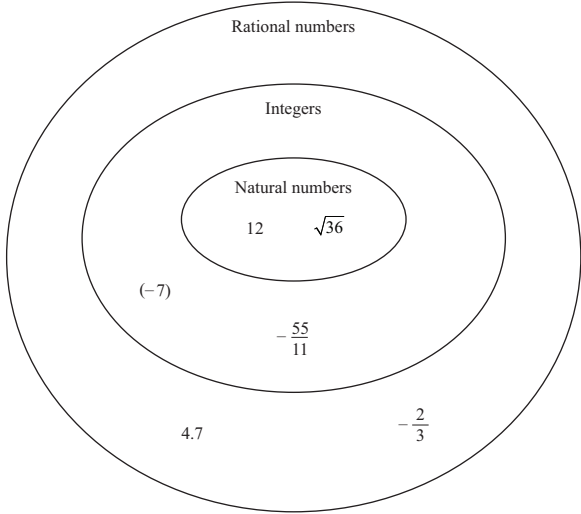
The table shows acceptable and unacceptable versions of various answers.

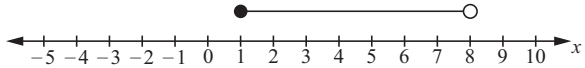
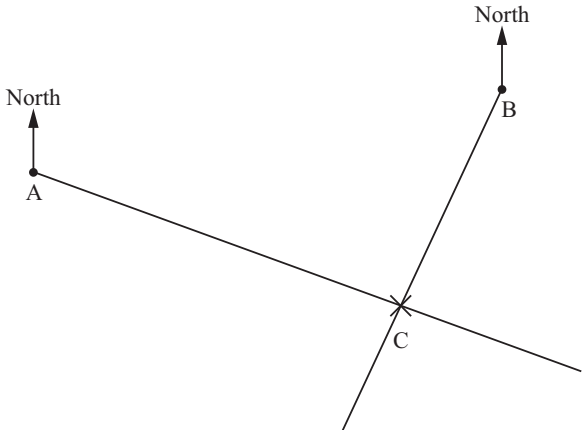
For the following	Accept	Do not accept
For $6 \leq x < 8$	$[6, 8)$	$< x <$
For $x \leq -2$	$(-\infty, -2]$	$x < -2$
For $x > 3$	$(3, \infty)$ $3 < x$	Just '3' written on the answer line, even if $x > 3$ appears in the working

Plotting points

The table shows acceptable and unacceptable ways to plot points.

Accept	Do not accept
Crosses or dots plotted within $\pm \frac{1}{2}$ square of the correct answer	A horizontal line and vertical line from the axes meeting at the required point
The graph line passing through a point implies the point even though there is no cross	

Question	Answer	Marks	Part Marks	Guidance
1	5	1		
2(a)	38.5 and 37	1		Both answers in the correct order for the mark.
2(b)	-0.5	1		
3(a)	7	1		
3(b)	0.85 (m)	1		
4(a)		2	Award 1 mark for three or four correctly inserted values.	
4(b)	irrational	1		Accept any clear indication.
5	$x^2 + 9x + 14$ final answer	2	Award 1 mark for three correct terms from $x^2 + 7x + 2x + 14$	Note $9x$ counts as two terms.

Question	Answer	Marks	Part Marks	Guidance
6(a)	$1 \leq x < 8$	1		
6(b)		1		Follow through <i>their</i> inequality from part (a) if in the form $a \leq x < b$ or $a < x \leq b$
7	Correct position for C drawn on the map.  and 10.25 to 12.5 (m)	3	Award 2 marks for correct position of C. or Award 1 mark for one bearing correct for C. and If 1 or 0 mark(s) for position of C, award 1 mark for <i>their</i> $BC \times 250 \div 100$	Tolerance $\pm 2^\circ$ Tolerance $\pm 2^\circ$ Accept tolerance of ± 2 mm on <i>their</i> BC.
8(a)	Ruled straight line extending at least as far as (0, 1) and (4, 5).	2	Award 1 mark for a line with gradient 1 or y-intercept at 1	
8(b)	(1.5, 2.5) or equivalent	1		Follow through coordinates of <i>their</i> point of intersection.

Question	Answer	Marks	Part Marks	Guidance
9(a)	4200 (mg)	1		
9(b)	3.5×10^{-4} (g)	2	Award 1 mark for $4.2 \div 12\,000$ or figs 35 or for <i>their</i> decimal answer written correctly in standard form.	Figs 35 could be, e.g. 0.035 or 35 000 etc.
10	(\$) 2.50 and (\$) 0.90	2	Award 1 mark for (\$) 2.50 or (\$) 0.90	Accept answers in either order. Accept 2.5 etc.
11(a)	$\frac{5}{6}$ in all three spaces	1		All three spaces correct for the mark.
11(b)	$\frac{5}{18}$ or equivalent	2	Award 1 mark for $\frac{1}{6} \times \frac{5}{6} (\times 2)$ or equivalent. or for correct answer from $\frac{1}{6} \times \textit{their}$ $\frac{5}{6} \times 2$ provided <i>their</i> $\frac{5}{6} < 1$	For 2 marks, accept equivalent decimal or percentage. 1 mark implied by an answer of $\frac{5}{36}$ or equivalent.
11(c)	independent	1		Accept any clear indication.

Question	Answer	Marks	Part Marks	Guidance
12	84.4 (cm ²)	3	Award 2 marks for 84.35 to 84.43... or Award 1 mark for $(0.5 \times) \pi \times 7.33^2$ and award 1 mark for <i>their</i> more accurate answer correctly rounded to three sig figs.	Accept value of π between 3.14 and $\frac{22}{7}$
13	Elk ticked and $(72.4 \times \frac{5}{8})$ or equivalent) = 45.25 (miles/hour) or Elk ticked and $(43.0 \times \frac{8}{5})$ or equivalent) = 68.8 (kilometres/hour)	2	Award 1 mark for 5 miles = 8 kilometres seen or implied, e.g. $72.4 \times \frac{5}{8}$ or equivalent or $43.0 \times \frac{8}{5}$ or equivalent	For 2 marks, accept 45, 45.3 or 69 Accept more accurate conversion values used, e.g. 1.61, 1.609...

Question	Answer	Marks	Part Marks	Guidance
14(a)	The sample is small. and An explanation implying the sample may not be representative (of the population).	2	Award 1 mark for The sample is small. or An explanation implying the sample may not be representative (of the population).	Accept any explanation suggesting the sample may not be representative, e.g. - Boys from the running club may be unusually tall. - The boys and girls are not taken from the same sub-group.
14(b)	A correct explanation stating that the intervals do not include all heights.	1		Accept, e.g. 150/160/170 (cm) is missing. - There is nowhere to put 150 (cm).
14(c)	Three class intervals that include every value once only, e.g. $(h < 150)$, $150 \leq h < 160$, $160 \leq h < 170$, $h \geq 170$	1		All three intervals correct for the mark. Accept unequal class intervals provided they include every value once only.
15(a)	B with vertices (1, 1), (4, 1), (4, 2), (2, 2), (2, 3) and (1, 3) and C with vertices (1, -5), (4, -5), (4, -6), (2, -6), (2, -7) and (1, -7)	2	Award 1 mark for B with vertices (1, 1), (4, 1), (4, 2), (2, 2), (2, 3) and (1, 3) or Correct reflection of <i>their</i> B in $y = -2$	
15(b)	180° rotation and centre (3, -2)	2	Award 1 mark for 180° rotation or centre (3, -2)	Accept follow through provided single transformation possible.

Question	Answer	Marks	Part Marks	Guidance
16(a)	$(v =) 10.5$ and $(v =) -10.5$	2	Award 1 mark for 10.5 or -10.5 or $v^2 = 110.25$	
16(b)	$(a =) \frac{v^2 - u^2}{2s}$ or $(a =) \frac{v^2}{2s} - \frac{u^2}{2s}$ final answer	2	Award 1 mark for $2as = v^2 - u^2$ or $a + \frac{u^2}{2s} = \frac{v^2}{2s}$	Do not accept, e.g. $\frac{\frac{v^2 - u^2}{2}}{s}$ for 2 marks.
17	116(m)	3	Award 2 marks for $(CD =) \sqrt{24^2 + (42 - 10)^2}$ or better. or Award 1 mark for $BC^2 + 24^2 = 26^2$ or better or $CD^2 = 24^2 + (42 - \text{their } BC)^2$ or better.	Or better, e.g. $CD = 40$ Or better, e.g. $\sqrt{26^2 - 24^2}$ or $BC = 10$ Or better, e.g. $\sqrt{24^2 + (42 - \text{their } BC)^2}$

Question	Answer	Marks	Part Marks	Guidance
18		3	Award 2 marks for all five correct but not simplified or for three or four correct (accept unsimplified and allow follow through as detailed below). or Award 1 mark for one or two correct (accept unsimplified and allow follow through as detailed below). For follow through:	'c' means <i>their c</i>, etc.

